

LEORS

LIVE EVENT OPERATIONAL READINESS STANDARD

Comprehensive Operational Playbook

Mapped to NIN Methodology & Sentrais OS
Temporal Orchestration Framework

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Sentrais Corporation

"Calm during chaos. Making the complex simple."

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1. Executive Summary

LEORS (Live Event Operational Readiness Standard) establishes a certifiable, insurance-legible, regulator-friendly standard for multi-stakeholder readiness and live operations performance. This playbook maps LEORS to the Sentrais OS architecture, NIN (NOVATE Intelligence Network) methodology, and provides temporal orchestration specifications for real-time operational experience.

1.1 Strategic Positioning

LEORS is the entertainment equivalent of an aviation-style safety and readiness regime: evidence-based, repeatable, auditable, and designed to prevent coordination failure under pressure. Unlike traditional compliance frameworks that check boxes, LEORS certifies system capability—the ability of an event ecosystem to reliably execute across governance, operations, safety, and learning dimensions.

1.2 This Playbook Delivers

- Complete NIN 5D methodology mapping (Discover → Diagnose → Design → Deploy → Debrief)
- Temporal orchestration with EVERGAME lifecycle integration (PREPARE → READY → RUN → REVIEW)
- Real-time evidence capture workflows via Sentrais Evidence Ledger
- Control catalog with 90+ controls mapped to n8n workflow architecture
- Gap analysis identifying elements not in original LEORS specification

2. NIN Methodology Mapping

The NOVATE Intelligence Network (NIN) provides the forensic discovery and intelligence framework that powers LEORS certification. Each NIN phase maps directly to LEORS certification activities.

2.1 The 5D Framework Applied to LEORS

NIN Phase	LEORS Activity	Sentrais Component	Evidence Output
DISCOVER	Intake & Risk Classification	Blueprint360 venue mapping	Risk register, stakeholder matrix
DIAGNOSE	Pre-Audit Readiness Review	SIPE gap analysis engine	Coordination deficit score
DESIGN	Evidence Collection Window	n8n workflow configuration	Control artifact package
DEPLOY	Simulation & Go-Live Observation	Temporal simulation engine	Real-time telemetry capture
DEBRIEF	Scorecard, Findings & Renewal	Evidence Ledger + AAR	Certification issuance

2.2 DISCOVER Phase: Forensic Mapping

The DISCOVER phase establishes the complete operational landscape through NIN's forensic discovery methodology:

Stakeholder Ecosystem Mapping

- Primary operators: Venue management, production companies, promoters
- Public safety: Police, fire, EMS, emergency management
- Support services: Medical, security contractors, transportation
- External dependencies: Weather services, ticketing, broadcast

Sentrais Integration: Blueprint360

Blueprint360 creates the digital twin of the venue/event ecosystem, capturing physical layout, operational dependencies, decision rights, and communication pathways. This becomes the foundation for all LEORS controls.

2.3 DIAGNOSE Phase: Gap Analysis

DIAGNOSE identifies coordination deficits through systematic assessment:

Coordination Deficit Scoring

Deficit Category	Assessment Method	Score Range	LEORS Impact
Authority Clarity	Decision rights matrix	0-25	No-fail control

	review		
Communication Pathways	Protocol stress testing	0-25	Domain 3 weight
Resource Visibility	Integration health check	0-25	Domain 5 weight
Learning Integration	AAR process assessment	0-25	Domain 8 weight

Sentrais Integration: SIPE Engine

The Sentrais Intelligence Processing Engine (SIPE) runs predictive analysis on discovered data, identifying patterns of coordination failure risk before they manifest operationally.

3. Temporal Orchestration Framework

LEORS certification operates within Sentrais OS's temporal architecture, which manages the complete event lifecycle through precisely defined state machines and timing sequences.

3.1 EVERGAME Lifecycle Integration

The EVERGAME framework provides four operational zones that map directly to LEORS gates:

EVERGAME Zone	LEORS Gate	Temporal Window	Key Activities	Evidence Captured
PREPARE	Gate A: Design	T-180 to T-30 days	Blueprint creation, risk register, stakeholder alignment	Policy artifacts, planning documents
READY	Gate B: Advance	T-30 to T-1 day	Integration testing, simulation drills, final checks	System health, drill outcomes
RUN	Gate C: Activate / Gate D: Live	T-0 (gates open to facility clear)	Real-time operations, incident response, telemetry	Continuous evidence stream
REVIEW	Post-Certification	T+1 hour to T+30 days	AAR generation, pattern analysis, renewal	Lessons learned, improvement tasks

3.2 Real-Time State Machine

During RUN phase, the Sentrais state machine manages sub-phase transitions with LEORS evidence capture at each boundary:

RUN Phase Sub-States

- 1. pre_gates → Evidence: Staff positions verified, systems online
- 2. gates_open → Evidence: Crowd ingress telemetry begins
- 3. pre_event → Evidence: Final go/no-go captured
- 4. event_active → Evidence: Continuous monitoring (30s intervals)
- 5. halftime/intermission → Evidence: Crowd movement, concession load
- 6. event_active_2 → Evidence: Second half monitoring
- 7. post_event → Evidence: Egress flow, incident resolution
- 8. facility_clear → Evidence: Final counts, handoff to REVIEW

3.3 Time Acceleration for Simulation

The Temporal Simulation Engine enables LEORS certification testing at accelerated speeds:

Mode	Speed	Real Duration	Simulated	Use Case

			Duration	
realtime	1x	4 hours	4 hours	Full certification exercises
fast	10x	24 minutes	4 hours	QA testing
rapid	60x	4 minutes	4 hours	Control validation
demo	300x	1.5 minutes	~7.5 hours	Sales demonstrations

4. Evidence Ledger Architecture

The Sentrais Evidence Ledger provides the immutable audit trail required for LEORS certification. Every decision, action, and outcome is captured with cryptographic hash chains ensuring evidence integrity.

4.1 Evidence Types for LEORS

Evidence Type	Category	Source Workflow	LEORS Domain
state_transition	lifecycle	LIFECYCLE_*_Phase	All domains
health_check	system	INTEGRATION_System_Health_Monitor	Domain 5
incident_created	incident	RUN_Incident_Handler	Domain 4, 6
sop_activated	sop_execution	RESPONSE_SOP_Executor	Domain 3, 4
weather_threat	weather	RUN_Monitor_Weather	Domain 4
crowd_alert	crowd_safety	RUN_Monitor_Crowd	Domain 4
compliance_validation	compliance	REPORTING_Compliance_Report	Domain 1
go_nogo_decision	governance	LIFECYCLE_Ready_Phase	Domain 1, 2

4.2 Evidence Strength Multiplier (ESM)

LEORS scoring incorporates evidence quality through the ESM system:

Evidence Type	Example	ESM	Tier Requirement
Self-attested	PDF checklists, signed forms	0.90	Bronze minimum
Supervisor verified	Signed digital logs	1.00	Bronze critical
System recorded	n8n workflow execution logs	1.10	Silver minimum
Immutable ledger	Evidence Ledger with hash chain	1.15	Gold minimum
Telemetry validated	Sensor + system correlation	1.20	Platinum requirement

5. Control Framework

5.1 Control Type Taxonomy

Type	Definition	Tier Focus	Sentrais Implementation
P (Policy)	Documented, approved, versioned	Bronze core	MongoDB document store
O (Operational)	Executed via workflow/checklist	Bronze/Silver	n8n workflow execution
E (Evidence)	Immutable completion record	All tiers	Evidence Ledger entry
T (Telemetry)	Measurable real-time signal	Gold/Platinum	TimescaleDB time-series

5.2 No-Fail Critical Controls

The following controls are non-negotiable. Any failure blocks certification regardless of overall score:

9. **LEORS-GOV-01: Authority clarity for cancellation and stop-show decisions**
10. **LEORS-CMD-01: Unified incident escalation protocol with time thresholds**
11. **LEORS-CRD-01: Ingress/egress capacity plan validated against ticketed volume**
12. **LEORS-MED-01: Medical surge plan and triage routing**
13. **LEORS-COM-01: Communications redundancy plan**
14. **LEORS-CYB-01: Cyber incident response for ticketing and entry systems**
15. **LEORS-WTH-01: Weather monitoring and action thresholds**
16. **LEORS-CRD-02: Credentialing integrity rules for restricted zones**

5.3 Domain Control Mapping to n8n Workflows

LEORS Domain	Primary n8n Workflows	Control Count
1. Governance & Decision Rights	LIFECYCLE_Prepare_Phase, SHARED_State_Machine	12
2. Event Readiness Lifecycle	LIFECYCLE_Ready_Phase, PREPARE_* workflows	14
3. Command & Coordination	RUN_Incident_Handler, RESPONSE_Escalation	15
4. Crowd & Life Safety	RUN_Monitor_Crowd, RESPONSE_SOP_Executor	18
5. Production & Infrastructure	RUN_Monitor_Systems, INTEGRATION_*	10
6. Medical & Emergency	RESPONSE_SOP_Executor	8

Response	(medical SOPs)	
7. Cyber, Data & Privacy	INTEGRATION_System_Health_Monitor	8
8. After-Action Learning	LIFECYCLE_Review_Phase, REPORTING_After_Action	7

6. Identified Gaps & Recommended Enhancements

The following elements were not explicitly addressed in the original LEORS specification but are critical for comprehensive operational readiness:

6.1 Missing from Original Specification

A. Artist/Production Integration Layer

Original LEORS focuses on venue and public safety stakeholders but lacks specific controls for artist management teams, production companies, and touring operations. Recommended addition:

- LEORS-PRD-01: Production rider compliance verification workflow
- LEORS-PRD-02: Artist cancellation authority protocol (VFLA v. WME alignment)
- LEORS-PRD-03: Load-in/out timing variance tracking

B. Financial Impact Tracking

LEORS captures operational evidence but lacks connection to financial outcomes:

- LEORS-FIN-01: Curfew penalty exposure monitoring
- LEORS-FIN-02: Insurance claim evidence package automation
- LEORS-FIN-03: Per-event profitability correlation to readiness score

C. Fan Experience Integration

Safety and operations are covered, but fan satisfaction metrics affect business outcomes:

- LEORS-FAN-01: Ingress time measurement against NPS correlation
- LEORS-FAN-02: Concession queue abandonment tracking
- LEORS-FAN-03: VIP experience delivery verification

D. Vendor Ecosystem Certification

LEORS certifies venues/events but not the vendor ecosystem:

- LEORS-VND-01: Security contractor certification requirements
- LEORS-VND-02: Medical service provider capability validation
- LEORS-VND-03: Technology vendor integration testing protocol

E. Mental Health & Workforce Wellbeing

Post-COVID industry shift requires attention to workforce factors:

- LEORS-WRK-01: IMSAFE-equivalent crew readiness assessment
- LEORS-WRK-02: Maximum shift duration enforcement
- LEORS-WRK-03: Critical role fatigue monitoring

6.2 Recommended Tier Adjustments

Tier	Original Requirement	Enhanced Requirement	Rationale
Bronze	Manual evidence package	Structured digital forms	Reduces assessor burden
Silver	Telemetry in 2 domains	Telemetry in 3 domains + API health	System reliability baseline
Gold	Unified command evidence	+ Protected dissent capture	Prevents Challenger-style failures
Platinum	Predictive + federated	+ Cross-tour intelligence sharing	Network effect realization

7. Certification Process

7.1 Certification Lifecycle

17. Intake and Risk Classification (Week 1)

NIN DISCOVER phase executes. Venue profile created in Blueprint360. Risk tier assigned (R1-R5).

18. Pre-Audit Readiness Review (Week 2-3)

NIN DIAGNOSE phase. SIPE gap analysis. Coordination deficit score calculated. Remediation plan issued.

19. Evidence Collection Window (Week 4-8)

NIN DESIGN phase. n8n workflows configured. Artifact package assembled. Policy controls validated.

20. Simulation or Tabletop Validation (Week 9-10)

Temporal simulation engine runs certification scenarios. SOP execution tested. Telemetry validated.

21. Go-Live Observation (Optional Bronze, Required Gold+)

NIN DEPLOY phase. Assessor present for live event. Real-time evidence capture validated.

22. Scorecard and Findings (Week 11)

NIN DEBRIEF phase. Domain scores calculated. ESM applied. Tier threshold validated.

23. Certification Issuance

Certificate generated with QR code linking to Evidence Ledger. Published to LEORS registry.

24. Renewal and Continuous Monitoring

Cadence varies by tier (see Section 7.2). Evidence Ledger continuously updated.

7.2 Renewal Cadence

Tier	Renewal Cycle	Interim Requirements
Bronze	12 months	None
Silver	12 months	Mid-year attestation
Gold	12 months	Quarterly signal checks
Platinum	Annual recertification	Continuous monitoring

8. Productization Options

8.1 Service Tiers

Option A: Certification-as-a-Service

Annual certification + audit + renewal. Target: Independent venues, small promoters.

Pricing: \$25K-\$75K per certification cycle

Option B: Certification + Managed Readiness

Monthly readiness ops, quarterly drills, evidence maintenance. Target: Arena operators, regional chains.

Pricing: \$85K-\$150K annually

Option C: Tour-Ready Certification

"These venues are LEORS Gold for this tour profile." Target: Artist management, touring companies.

Pricing: \$150K-\$250K per tour

Option D: Enterprise Control Tower

Multi-venue standardization, cross-portfolio benchmarking, federated learning. Target: Live Nation, AEG.

Pricing: \$2M-\$4M annually

8.2 Strategic Value Proposition

Stakeholder	LEORS Value	Commercial Lever
Venues	Reduced liability, insurance leverage	Premium pricing for events
Promoters	Risk insulation, operational predictability	Lower insurance premiums
Artists	Brand protection, safety assurance	Venue selection criteria
Insurers	Evidence-based underwriting	Premium differentiation
Regulators	Voluntary compliance standard	Reduced enforcement burden

9. Appendices

Appendix A: Complete Control Catalog Structure

Each control in the LEORS catalog follows this schema:

- Control ID: LEORS-[DOMAIN]-[NUMBER] (e.g., LEORS-GOV-01)
- Control Type: P (Policy), O (Operational), E (Evidence), T (Telemetry)
- Tier Requirement: Bronze / Silver / Gold / Platinum
- Evidence Required: Specific artifact or data stream
- Sentrais Workflow: n8n workflow that validates/captures control
- ESM Range: Evidence strength multiplier applicable
- No-Fail Flag: Boolean indicating critical control status

Appendix B: Sentrais Database Schema Alignment

LEORS evidence maps to the following Sentrais data stores:

- PostgreSQL (Operational): events, incidents, sop_executions, compliance_reports
- TimescaleDB (Time-Series): sensor_readings, crowd_metrics, weather_readings
- MongoDB (Documents): sops, after_action_reports, compliance_frameworks
- Redis (Cache/State): dashboard snapshots, session state, circuit breakers
- Evidence Ledger (Immutable): All certification evidence with hash chains

Appendix C: Risk Profile Weighting Adjustments

Profile	Governance	Command	Crowd/Life Safety	Cyber
R1: Club/Theater	15%	15%	15%	10%
R2: Arena	15%	18%	15%	12%
R3: Stadium	15%	20%	18%	15%
R4: Festival	12%	15%	22%	10%
R5: SEAR-level	20%	20%	18%	18%

— END OF PLAYBOOK —

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